CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 4: Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Word problems

1. Over a 200 km journey, a fast train takes one hour less than a slow train. The slow train runs 10 km/h slower than the fast one. Find the speed of each train. CBSE 2026, 5 marks

SOLUTION

Let the fast train run at x km/h. Then the slow train runs at $(x-10)$ km/h.

$$\frac{200}{x-10}-\frac{200}{x}=1$$

$200x-200(x-10)=x(x-10)$, so $2000=x^2-10x$

$x^2-10x-2000=0$, so $(x-50)(x+40)=0$

$x=50$ or $x=-40$. A speed cannot be negative.

Check: $200\div 50=4$ hours and $200\div 40=5$ hours, which is one hour more.

Answer: 50 km/h and 40 km/h

2. Two squares have a total area of 640 m², and their perimeters differ by 64 m. Find the side of each square. CBSE 2026, 5 marks

SOLUTION

Let the sides be x m and y m with $x>y$.

$4x-4y=64$, so $x=y+16$.

$x^2+y^2=640$: $(y+16)^2+y^2=640$, so $2y^2+32y-384=0$

$y^2+16y-192=0$, so $(y+24)(y-8)=0$

$y=8$ (a length cannot be negative) and $x=24$.

Check: $24^2+8^2=576+64=640$.

Answer: 24 m and 8 m

3. A right-angled triangle has a perimeter of 60 cm and a hypotenuse of 25 cm. Find the lengths of the other two sides. CBSE 2025, 5 marks

SOLUTION

Let one of the other sides be x cm. The two sides add up to $60 - 25 = 35$ cm, so the other side is $(35 - x)$ cm.

By the Pythagoras theorem, $x^2 + (35 - x)^2 = 25^2$.

$x^2 + 1225 - 70x + x^2 = 625$, so $2x^2 - 70x + 600 = 0$, that is $x^2 - 35x + 300 = 0$.

$(x - 15)(x - 20) = 0$, so $x = 15$ or $x = 20$.

If $x = 15$ the other side is 20, and if $x = 20$ the other side is 15.

Answer: 15 cm and 20 cm

4. A train covers 480 km at a uniform speed. Had it travelled 8 km/h slower, the journey would have taken 3 hours longer. Find the speed of the train. CBSE 2025, 5 marks

SOLUTION

Let the speed be x km/h. The usual time is $\frac{480}{x}$ hours and the slower time is $\frac{480}{x - 8}$ hours.

$\frac{480}{x - 8} - \frac{480}{x} = 3$

$480x - 480(x - 8) = 3x(x - 8)$, so $3840 = 3x^2 - 24x$

$x^2 - 8x - 1280 = 0$, so $(x - 40)(x + 32) = 0$

$x = 40$ or $x = -32$. A speed cannot be negative.

Check: $480 \div 40 = 12$ hours and $480 \div 32 = 15$ hours, which is 3 hours more.

Answer: 40 km/h

5. A rectangular floor is covered exactly by 200 square tiles. With tiles whose side is 1 unit longer, only 128 tiles are needed. Let the side of an original tile be x units. (i) Form a quadratic equation in x. (ii) Write it in standard form. (iii) Find x by factorisation, OR solve the equation by the quadratic formula. CBSE 2024, 4 marks, case study

SOLUTION

(i) The floor area is the same in both cases: $200x^2 = 128(x + 1)^2$

(ii) $200x^2 = 128x^2 + 256x + 128$, so $72x^2 - 256x - 128 = 0$. Dividing by 8: $9x^2 - 32x - 16 = 0$

(iii) By factorisation: $9x^2 - 36x + 4x - 16 = 0$, so $9x(x - 4) + 4(x - 4) = 0$ and $(9x + 4)(x - 4) = 0$

$x = 4$ or $x = -\frac{4}{9}$. A length cannot be negative, so $x = 4$.

OR by the formula: $x = \frac{32 \pm \sqrt{1024 + 576}}{18} = \frac{32 \pm 40}{18}$, which gives $x = 4$ or $x = -\frac{4}{9}$.

Answer: the side of a tile is 4 units.

6. A school photo is 18 cm long and 12 cm wide. Its length and width are each to be increased by x cm so that the area doubles. (i) Write an equation for this. (ii) Write it as a quadratic equation in standard form. (iii) Find the new length and width, OR decide whether any rational value of x gives a new area of 220 cm². CBSE 2023, 4 marks

SOLUTION

(i) Original area $= 18 \times 12 = 216$ cm², so the new area is 432 cm². $(18 + x)(12 + x) = 432$

(ii) $216 + 30x + x^2 = 432$. $x^2 + 30x - 216 = 0$

(iii) $(x + 36)(x - 6) = 0$, so $x = 6$ (a length cannot be negative). **New size: 24 cm by 18 cm.**

OR: $(18 + x)(12 + x) = 220$ gives $x^2 + 30x - 4 = 0$, with $D = 900 + 16 = 916$. This is not a perfect square, so the roots are irrational. **No rational value of x works.**

7. Rehan's age 5 years ago multiplied by his age 7 years from now is one more than twice his present age. Find his present age. CBSE 2022 Term 2, 2 marks

SOLUTION

Let his present age be x years. Then $(x - 5)(x + 7) = 2x + 1$.

$x^2 + 2x - 35 = 2x + 1$, so $x^2 = 36$ and $x = \pm 6$.

An age cannot be negative.

Answer: 6 years

8. Two taps together fill a tank in $1\frac{7}{8}$ hours. Working alone, the larger tap takes 2 hours less than the smaller one. How long does each tap take on its own? CBSE 2019, 4 marks

SOLUTION

Let the larger tap take x hours. Then the smaller tap takes $x + 2$ hours, and together they take $\frac{15}{8}$ hours.

In one hour: $\frac{1}{x} + \frac{1}{x+2} = \frac{8}{15}$

$15(2x + 2) = 8x(x + 2)$, so $8x^2 - 14x - 30 = 0$, that is $4x^2 - 7x - 15 = 0$

$(4x + 5)(x - 3) = 0$, so $x = 3$ (time cannot be negative).

Answer: the larger tap takes 3 hours and the smaller tap takes 5 hours

9. A plane took off 30 minutes late. To reach its destination 1500 km away on time, it flew 100 km/h faster than usual. Find its usual speed. CBSE 2018, 3 marks

SOLUTION

Let the usual speed be x km/h. The faster flight saves half an hour.

$\frac{1500}{x} - \frac{1500}{x+100} = \frac{1}{2}$

$1500 \times 100 \times 2 = x(x + 100)$, so $x^2 + 100x - 300000 = 0$

$(x + 600)(x - 500) = 0$, so $x = 500$ (speed cannot be negative).

Answer: 500 km/h

10. A motor boat has a speed of 18 km/h in still water. It takes 1 hour longer to go 24 km upstream than to come back the same distance downstream. Find the speed of the stream. CBSE 2018, 4 marks

SOLUTION

Let the speed of the stream be x km/h. Upstream speed $= 18 - x$ and downstream speed $= 18 + x$.

$$\frac{24}{18-x} - \frac{24}{18+x} = 1$$

$24(18+x) - 24(18-x) = 324 - x^2$, so $48x = 324 - x^2$ and $x^2 + 48x - 324 = 0$

$(x+54)(x-6) = 0$, so $x = 6$.

Answer: 6 km/h

11. A train covers 63 km at a certain average speed and then 72 km at a speed 6 km/h faster. The whole journey takes 3 hours. Find the first speed. CBSE 2018, 4 marks

SOLUTION

Let the first speed be x km/h. $\frac{63}{x} + \frac{72}{x+6} = 3$

$63(x+6) + 72x = 3x(x+6)$, so $135x + 378 = 3x^2 + 18x$

$3x^2 - 117x - 378 = 0$, that is $x^2 - 39x - 126 = 0$

$(x-42)(x+3) = 0$, so $x = 42$.

Answer: 42 km/h

12. A motor boat has a speed of 24 km/h in still water. It takes 1 hour longer to go 32 km upstream than to return the same distance downstream. Find the speed of the stream. CBSE 2016, 4 marks

SOLUTION

Let the speed of the stream be x km/h.

$$\frac{32}{24-x} - \frac{32}{24+x} = 1$$

$32(24+x) - 32(24-x) = 576 - x^2$, so $64x = 576 - x^2$ and $x^2 + 64x - 576 = 0$

$(x+72)(x-8) = 0$, so $x = 8$.

Answer: 8 km/h

2 Nature of roots

13. The equation $ax^2 + bx + c = 0$, $a \neq 0$, has real and equal roots. Which relation holds? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ CBSE 2024, 1 mark

SOLUTION

For real and equal roots, $D = b^2 - 4ac = 0$.

So $b^2 = 4ac$, that is $ac = \frac{b^2}{4}$.

Answer: (c)

14. What is the least positive value of k for which $2x^2 + kx - 4 = 0$ has rational roots? (a) $\pm 2\sqrt{2}$ (b) 2 (c) ± 2 (d) $\sqrt{2}$ CBSE 2023, 1 mark

SOLUTION

The roots are rational when the discriminant is a perfect square. $D = k^2 - 4(2)(-4) = k^2 + 32$.

Among the options, $k = \sqrt{2}$ gives $D = 34$, which is not a perfect square. $k = 2$ gives $D = 36 = 6^2$.

Answer: (b)

15. For what value of m does $(m-1)x^2 + 2(m-1)x + 1 = 0$ have two equal real roots? CBSE 2022 Term 2, 2 marks

SOLUTION

Equal roots need $D = 0$: $[2(m-1)]^2 - 4(m-1)(1) = 0$

$4(m-1)(m-1-1) = 0$, so $m = 1$ or $m = 2$.

If $m = 1$ the x^2 term disappears and the equation is no longer quadratic, so $m = 1$ is rejected.

Answer: $m = 2$

16. For what values of k does $x^2 + 4x + k = 0$ have real roots? CBSE 2019, 1 mark

SOLUTION

Real roots need $D \geq 0$: $16 - 4k \geq 0$

Answer: $k \leq 4$

17. The roots of $3x^2 - 10x + k = 0$ are reciprocals of each other. Find k. CBSE 2019, 1 mark

SOLUTION

If the roots are α and $\frac{1}{\alpha}$, their product is 1.

Product of roots $= \frac{k}{3} = 1$

Answer: $k = 3$

18. -5 is a root of $2x^2 + px - 15 = 0$, and the equation $p(x^2 + x) + k = 0$ has equal roots. Find k. CBSE 2016, 2 marks

SOLUTION

Put $x = -5$ in the first equation: $50 - 5p - 15 = 0$, so $p = 7$.

The second equation is $7x^2 + 7x + k = 0$. Equal roots need $D = 0$: $49 - 28k = 0$.

Answer: $k = \frac{7}{4}$

3 Solving by factorisation

19. Solve for x: $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$. CBSE 2022 Term 2, 2 marks

SOLUTION

Product of the first and last coefficients $= \sqrt{3} \times 7\sqrt{3} = 21 = 3 \times 7$, and $3 + 7 = 10$.

$\sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = 0$, so $\sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = 0$

$(x + \sqrt{3})(\sqrt{3}x + 7) = 0$

Answer: $x = -\sqrt{3}$ or $x = -\frac{7}{\sqrt{3}} = -\frac{7\sqrt{3}}{3}$

20. $x = 3$ is a root of $x^2 - 2kx - 6 = 0$. Find k. CBSE 2018, 1 mark

SOLUTION

Put $x = 3$: $9 - 6k - 6 = 0$, so $6k = 3$.

Answer: $k = \frac{1}{2}$

21. Solve for x: $\frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = \frac{2}{3}$, where $x \neq 1, 2, 3$. CBSE 2016, 3 marks

SOLUTION

$\text{LHS} = \frac{(x-3)+(x-1)}{(x-1)(x-2)(x-3)} = \frac{2(x-2)}{(x-1)(x-2)(x-3)} = \frac{2}{(x-1)(x-3)}$

$\frac{2}{(x-1)(x-3)} = \frac{2}{3}$, so $(x-1)(x-3) = 3$

$x^2 - 4x + 3 = 3$, so $x(x-4) = 0$

Answer: $x = 0$ or $x = 4$

4 Solving by the quadratic formula

22. Solve for x: $\frac{1}{x+1} + \frac{2}{x+2} = \frac{4}{x+4}$, where $x \neq -1, -2, -4$. CBSE 2016, 4 marks

SOLUTION

$\text{LHS} = \frac{(x+2)+2(x+1)}{(x+1)(x+2)} = \frac{3x+4}{x^2+3x+2}$

$(3x+4)(x+4) = 4(x^2+3x+2)$, so $3x^2 + 16x + 16 = 4x^2 + 12x + 8$

$x^2 - 4x - 8 = 0$. Here $D = 16 + 32 = 48$.

$x = \frac{4 \pm \sqrt{48}}{2} = \frac{4 \pm 4\sqrt{3}}{2}$

Answer: $x = 2 + 2\sqrt{3}$ or $x = 2 - 2\sqrt{3}$

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